

17 Use of the Data Booklet is relevant to this question.

Some relevant redox half-equations are given in the table.

half-equation	E° / V
$\text{I}_2(\text{aq}) + 2\text{e}^- \rightleftharpoons 2\text{I}^-(\text{aq})$	+0.54
$2\text{H}^+(\text{aq}) + \text{O}_2(\text{g}) + 2\text{e}^- \rightleftharpoons \text{H}_2\text{O}_2(\text{aq})$	+0.68
$\text{H}_2\text{O}_2(\text{aq}) + 2\text{H}^+(\text{aq}) + 2\text{e}^- \rightleftharpoons 2\text{H}_2\text{O}(\text{l})$	+1.77

What do the E° values shown in the table predict will be observed when a few drops of acidified aqueous hydrogen peroxide are added to an excess of aqueous potassium iodide?

- A The solution turns brown and effervescence occurs.
- B The solution turns brown without effervescence.
- C The solution turns purple and effervescence occurs.
- D The solution turns purple without effervescence.